CloudSage — Aftercare and Operations

What happens after the system is working

1. Continuous Operation

The production workflow should periodically discover resources, refresh cost/metric data, detect new optimization candidates, avoid duplicate recommendations and monitor previously applied optimizations.

2. Scheduled Jobs

- Daily resource discovery
- Daily/periodic optimization scan
- Cost refresh according to AWS data freshness
- Weekly recommendation review
- Post-change verification windows
- Periodic audit/report generation

3. Post-Change Verification

Every applied optimization creates a baseline. The system compares health, utilization, performance and cost after the change. If health degrades beyond policy thresholds, the optimization is marked failed and the rollback path is invoked according to the configured policy.

4. Rollback

Rollback should use the previous known-good Terraform configuration/plan where possible. Rollback itself should be controlled by the same approval and policy model unless a separately defined emergency policy exists.

5. Monitoring CloudSage

- API latency
- Worker failures
- AWS API errors/throttling
- Database health
- LLM latency/errors
- Token usage
- Recommendation throughput
- Approval latency
- Terraform failures
- Verification failures

6. Monitoring Optimization Quality

- Recommendations accepted/rejected
- False positives
- Predicted vs realized savings
- Failed optimizations
- Rollback frequency
- Average savings per accepted recommendation
- Time from detection to verification

7. Model Maintenance

When changing the LLM provider/model, run a fixed evaluation set before production use. Compare schema compliance, factual grounding, recommendation quality, safety behavior and cost.

8. AWS/API Maintenance

Keep boto3/AWS API usage current, handle API deprecations, maintain retry logic and review IAM permissions periodically.

9. Terraform Maintenance

Pin Terraform/provider versions where appropriate, test upgrades, inspect plan behavior and keep modules backward compatible.

10. Database Maintenance

- Indexes for resource lookups
- Retention policy for raw metrics
- Archive old audit data
- Backups
- Migration tooling
- Data integrity checks

11. Operational Principle

The system should fail closed for infrastructure writes: if evidence is incomplete, AI output is invalid, policy evaluation fails, Terraform plan is unexpected, or required approval is missing, the system must not apply the change.